Notes: Orr (JPE, 2022)

Within-Firm Productivity Dispersion: Estimates and Implications

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1 Big picture

- **Question.** How large is productivity dispersion *within* multiproduct firms, and can we identify product-line productivity when the data report inputs only at the firm or plant level rather than by product?
- **Core problem.** Standard production-function methods need product-line input use, but most manufacturing data sets observe only total labor, capital, and materials at the plant level. Without an input-allocation rule, differences in product-level output could reflect either input differences or true productivity differences.
- **Main theoretical contribution.** The paper derives a flexible mapping from observable output prices and quantities to unobserved input allocations across product lines. Under homogeneous product-line technologies and profit maximization, input shares are proportional to *output times marginal cost shares*. Once marginal costs are recovered from demand and pricing first-order conditions, product-level input use becomes identifiable.
- **Main empirical contribution.** The method is applied to a panel of Indian machinery plants from the 2000–2007 Annual Survey of Industries (ASI). The paper finds large within-plant heterogeneity: within-plant variation accounts for about 36% of product-level TFPQ dispersion and about 37% of product-level appeal dispersion.
- **Quantitative headline.** Removing a plant's worst product raises mean plant-level revenue efficiency by roughly 10% to just under 65%, with the largest gains for plants producing only two products.
- **Behavioral headline.** Plants are much more likely to drop weak products. A one-standard-deviation decline in product-level revenue efficiency raises the probability of dropping that product by about 6.5 percentage points.
- **Economic intuition.** Multiproduct plants have a "core competence." They are relatively good at some products and relatively bad at others. Competition and internal product selection then push plants to prune their lowest-performing lines.
- **Why this paper matters.** The paper shows that abstracting from within-firm heterogeneity is not innocuous. It matters for productivity measurement, product scope, product dropping, and how we think about multiproduct firms in trade, IO, and productivity analysis.

2 Data and measurement (key objects)

2.1 Primary data source and sample

- **Data.** Indian *Annual Survey of Industries* (ASI), 2000–2007.
- **Sampling frame.** Manufacturing plants with more than 10 workers. Plants with more than 100 workers are surveyed every year ("census" plants); smaller plants are sampled.
- **Panel structure.** The ASI contains plant identifiers that allow the author to form plant panels across years.
- **Industry focus.** The empirical application uses the machinery, equipment, and parts sector.

- **Why this sector?** It is relatively well suited to the paper's assumptions: compared with sectors like sugar and textiles, it is less affected by by-products, joint production, and obvious vertical integration across products.

2.2 Product, quantity, and price measurement

- **Product reporting.** Each plant reports revenues and quantities produced and sold for up to 10 products.
- **Product codes.** Products are classified using 5-digit ASICC codes. The machinery sample contains a little over 1,000 unique item codes.
- **Varieties versus product codes.** About 17% of plant-year observations contain multiple entries for the same 5-digit product code. The paper treats these as distinct *varieties* within the same product class rather than aggregating them.
- **Prices.** Product-level prices are unit values constructed from revenues and quantities sold, using the code-specific physical units attached to each ASICC code.
- **Dropped observations.** Product-year observations without quantity information (a bit over 12%) are excluded because the method needs quantity data.

2.3 Input measurement

- **Labor.** Measured by number of man-days worked.
- **Capital.** Constructed using the perpetual inventory method.
- **Materials.** Measured as materials expenditures deflated by a plant-level Cobb–Douglas input price index, which is meant to reduce input-price bias.
- **Unit of analysis.** The theoretical framework is stated at the firm-product level, but the ASI is a plant-level data set. The empirical analysis therefore treats plants as the relevant decision-making units.

2.4 Summary statistics

- **Plant-product level (Table 1).** Mean log revenue is 15.87, mean log quantity sold is 7.65, mean log price is 8.23, and mean log quantity produced is 7.78. About 76% of plant-product observations come from multiproduct plants.
- **Plant level (Table 2).** Mean log labor is 8.92, mean log capital is 15.15, mean log materials is 6.91, and the average number of varieties is 1.55. About 26% of plant-years are multiproduct in the restricted machinery sample.

3 Models and empirical specifications (all equations + interpretation)

3.1 General product-line production technology

For product j produced by plant i in year t, output is

$$Y_{it}^{j} = \exp\left(q_{it}^{j}\right) F\left(\mathbf{X}_{it}^{j}\right). \tag{1}$$

- **Interpretation.** Product-line output equals a Hicks-neutral productivity shifter times a common production technology applied to product-line inputs.
- **Key empirical object.** q_{it}^{j} is product-level log TFPQ.
- **Crucial restriction.** The shape of $F(\cdot)$ is identical across product lines within the plant; product lines differ only through Hicks-neutral productivity.

3.2 Assumptions behind the input-allocation result

The theorem rests on a set of assumptions that are worth stating clearly.

- **Homogeneity.** $F(\cdot)$ is homogeneous of degree $\phi > 0$, continuous, differentiable, strictly increasing, and quasi-concave.
- **Common technology shape.** Product lines differ only through q_{it}^{j}, not through different production-function parameters.
- **At least one static input.** The firm must use at least one static input (for example, materials) that can be purchased each period.
- **Costless transferability within the firm.** Inputs can be moved across product lines without extra within-firm adjustment costs.
- **Complete attribution.** Aggregate inputs can be allocated across product lines, so there are no public inputs in the baseline setup.
- **No internal vertical input-output links.** Products sold by the plant are not themselves inputs into other products sold by the same plant.
- **Demand system.** Demand satisfies connected substitutes, which ensures marginal-cost inversion from observed prices and quantities.
- **Optimization.** Plants choose prices, outputs, investments, aggregate inputs, and input allocations to maximize discounted profits.
- **Economic role of these assumptions.** They guarantee that prices reveal marginal costs and that marginal costs reveal product-line input shares.

3.3 Firm problem and conditional cost minimization

Let $\widetilde{\mathbf{P}}_t$ denote the vector of all product prices. The plant's dynamic problem can be written schematically as

$$V_t(x_{it}) = \max_{\widetilde{\mathbf{P}}_{it}, \mathbf{X}_{it}, \widetilde{\mathbf{X}}_{it}, \widetilde{\mathbf{I}}_{it}} \left\{ \sum_{j \in \mathcal{Y}_{it}} P_{it}^j Q_{it}^j(\widetilde{\mathbf{P}}_t, \widetilde{\mathbf{h}}_t) - C\Big(\widetilde{\mathbf{X}}_{it}, \widetilde{\mathbf{I}}_{it}\Big) + \beta \mathbb{E}_t\big[V_{t+1}(x_{i,t+1})\big] \right\}. \tag{2}$$

- **Interpretation.** The plant chooses product-level prices and input allocations to maximize firm value, taking demand, technology, and the evolution of dynamic inputs as given.
- **Why this matters.** The theoretical contribution comes from using the first-order conditions of this optimization problem to infer unobserved product-line inputs.

3.4 Input-allocation rule

Under the maintained assumptions, all input shares within product line j are proportional across inputs, so product-line input shares satisfy

$$S_{it}^j = \frac{\mathrm{MC}_{it}^j Y_{it}^j}{\sum_{k \in \mathcal{Y}_{it}} \mathrm{MC}_{it}^k Y_{it}^k}. \tag{3}$$

- **Interpretation.** A product gets a larger share of labor, materials, and capital when it generates a larger share of the plant's total variable cost bill, namely marginal cost times quantity.
- **Big insight.** Once MC_{it}^j is recovered from pricing behavior, the allocation of each input across product lines becomes observable to the econometrician.

3.5 Pricing first-order conditions and marginal-cost inversion

For Bertrand pricing, each product satisfies

$$Q_{it}^j + \sum_{k \in \mathcal{Y}_{it}} \frac{\partial Q_{it}^k}{\partial P_{it}^j} \left(P_{it}^k - \mathrm{MC}_{it}^k\right) = 0. \tag{4}$$

Stacking the first-order conditions across products yields the matrix relation

$$\mathbf{Q}_t + \Delta_t \left(\mathbf{P}_t - \mathbf{MC}_t\right) = \mathbf{0}, \tag{5}$$

where Δ_t collects demand derivatives weighted by the ownership matrix. Hence

$$\mathbf{MC}_t = \mathbf{P}_t + \Delta_t^{-1} \mathbf{Q}_t. \tag{6}$$

- **Interpretation.** Observed prices and quantities, together with estimated demand derivatives and ownership, reveal marginal costs.
- **Identification logic.** Demand estimation is the first step because the demand derivatives in Δ_t are needed to invert prices into marginal costs.
- **Connected substitutes.** This assumption guarantees that Δ_t is invertible under the relevant conduct assumptions.

3.6 Special case: revenue shares as the input-allocation rule

If the plant charges the same markup on all products, then $\mathrm{MC}_{it}^j = P_{it}^j / m_{it}$, so the common markup cancels and

$$S_{it}^j = \frac{P_{it}^j Y_{it}^j}{\sum_{k \in \mathcal{Y}_{it}} P_{it}^k Y_{it}^k}. \tag{7}$$

- **Interpretation.** Under equal markups, product-line input shares equal revenue shares.
- **Why this matters.** This result rationalizes the common empirical practice of allocating inputs by revenue shares, but only under restrictive conditions: no within-firm markup differences and common technology shape.

3.7 Discussion of extensions in the theory section

The paper also studies what happens when baseline assumptions fail.

- **General functional forms.** If production functions differ by product line or are not homogeneous, input shares solve a more complicated fixed-point system that depends on output elasticities. Then input allocation and production-function estimation become entangled.
- **Public or joint inputs.** With Cobb–Douglas and a special form of public inputs, the missing common-input component acts like an extra productivity shifter that depends on product scope.
- **Violating costless transferability.** If some dynamic inputs are specific to product lines, the researcher can sometimes treat them as part of the productivity residual, but then the residual contains endogenous specialized investments.
- **Takeaway.** The baseline theorem is powerful precisely because the assumptions make input use identifiable *without* needing to estimate output elasticities first.

3.8 Empirical product-line production function

In the application, the plant-product production technology is Cobb–Douglas:

$$Y_{it}^j = \exp\left(q_{it}^j\right) \left(L_{it}^j\right)^{\beta_L} \left(K_{it}^j\right)^{\beta_K} \left(M_{it}^j\right)^{\beta_M}. \tag{8}$$

- **Interpretation.** Product-line TFPQ is the residual after allocating labor, capital, and materials across products and partialling out their contributions.
- **Unknowns.** The empirical challenge is that L_{it}^j, K_{it}^j, and M_{it}^j are not observed in the ASI and must be recovered from demand-side information.

3.9 Demand system used in the application

The empirical demand system is a nested-logit-style specification over varieties within 5-digit product nests and 3-digit markets. The mean utility of variety j is

$$\delta_{it}^{j} = h_{it}^{j} - ap_{it}^{j}, \tag{9}$$

where h_{it}^{j} is product appeal and p_{it}^{j} is log price. The estimating equation implied by the demand system is

$$rs_{it}^{j} - rs_{h(j)t}^{0} = (1-\lambda) rs_{it}^{j|g(j)} - ap_{it}^{j} + h_{it}^{j}, \tag{10}$$

where rs_{it}^{j} is the log revenue share of product j in its 3-digit market, $rs_{h(j)t}^{0}$ is the log outside-option share, and $rs_{it}^{j|g(j)}$ is the log revenue share of the variety within its 5-digit product code.

- **Interpretation.** a is the price sensitivity parameter and $\lambda \in (0,1)$ controls within-nest substitution.
- **Why revenue shares?** The inversion is written in terms of revenue shares rather than quantity shares because plant-level quantity data are noisier and more vulnerable to unit mistakes.
- **Market definition.** The paper treats 3-digit ASICC sectors as markets and 5-digit product codes as nests within those markets.

3.10 Demand-side instruments

The main input-price-shifter instrument at the product-code level is

$$Z_t^{g(j)} = \sum_{k \in \mathcal{I}_g} \gamma_{kg} \log\left(\frac{1}{|\mathcal{F}_t^{kg}|} \sum_{m \in \mathcal{F}_t^{kg}} W_{mt}^{k} \right) - \frac{1}{T} \sum_{\tau=1}^{T} \sum_{k \in \mathcal{I}_g} \gamma_{kg} \log\left(\frac{1}{|\mathcal{F}_\tau^{kg}|} \sum_{m \in \mathcal{F}_\tau^{kg}} W_{m\tau}^{k} \right), \tag{11}$$

and the within-plant output-set instrument is

$$Z_{it}^{2jg} = \frac{\sum_{k \in \mathcal{Y}_{it}^{M}} Z_t^{g(k)} - Z_t^{g(j)}}{|\mathcal{Y}_{it}^{M}| - 1}. \tag{12}$$

- **Interpretation.** The first instrument uses input-price growth in other industries that buy similar materials. The second uses the plant's output mix to map those cost shocks into product-level pricing variation.
- **Exclusion restriction.** These shifters must affect a product's price through costs, not through demand or appeal shocks in machinery itself.

3.11 Productivity law of motion and quasi-differenced estimation

Productivity follows an AR(1) process:

$$q_{it}^{j} = \rho_0^{g(j)} + \rho_{d(j)} q_{i,t-1}^{j} + \nu_{it}^{j}, \tag{13}$$

where the persistence parameter varies by 2-digit industry. Using this law of motion, the production function is estimated from the quasi-differenced equation

$$
\begin{aligned}
y_{it}^{j} = \rho_0^{g(j)} + \rho_{d(j)} y_{i,t-1}^{j} + \beta_L\big(\widehat{\ell}_{it}^{j} - \rho_{d(j)}\widehat{\ell}_{i,t-1}^{j}\big) + \beta_K\big(\widehat{k}_{it}^{j} - \rho_{d(j)}\widehat{k}_{i,t-1}^{j}\big) \\
+ \beta_M\big(\widehat{m}_{it}^{j} - \rho_{d(j)}\widehat{m}_{i,t-1}^{j}\big) + \nu_{it}^{j} + \varepsilon_{it}^{j}.
\end{aligned} \tag{14}
$$

Moment conditions take the form

$$
\mathbb{E}\left[\left(\nu_{it}^{j} + \varepsilon_{it}^{j}\right)\mathbf{Z}_{it}^{j}\right] = \mathbf{0}. \tag{15}
$$

- **Interpretation.** The production function is estimated after replacing unobserved input use with the input allocations obtained from the first-step demand system.
- **Instruments.** The GMM routine uses lagged plant-level inputs, current capital under a predetermined-capital assumption, lagged output, and in some specifications the input-price instruments.
- **Preferred specification.** The paper prefers the specification that identifies the materials elasticity using the cost-shifter instruments rather than lagged materials, because the lagged-materials specification implies suspiciously large materials elasticities and increasing returns.

3.12 Revenue production function and product-level revenue efficiency

To measure plant-level efficiency in comparable units across product codes, the paper works with a revenue production function. Product-level log revenues satisfy

$$
\log R_{it}^{j} = \frac{a/\lambda}{1 + a/\lambda}\left(\beta_L \ell_{it}^{j} + \beta_K k_{it}^{j} + \beta_M m_{it}^{j}\right) + \widetilde{q}_{it}^{j} + \widetilde{h}_{it}^{j}, \tag{16}
$$

where the product-level residual is defined as

$$
E_{it}^{j} \equiv \widetilde{q}_{it}^{j} + \widetilde{h}_{it}^{j}. \tag{17}
$$

- **Interpretation.** E_{it}^{j} is *revenue efficiency*: a revenue-based productivity object that combines cost-side efficiency and demand-side product appeal.
- **Why not plain** TFPR**?** The paper emphasizes that this revenue residual is not the same as TFPQ$\times$ price. It nets out some price-quantity adjustment implied by the demand system.
- **Why use revenue efficiency?** Product-level TFPQ alone is hard to compare across product codes because physical units and unobserved quality differ. Revenue efficiency is in common revenue units and is more suitable for plant-level aggregation.

3.13 Plant-level revenue efficiency and the role of specialization

Plant-level log revenue efficiency decomposes as

$$
E_{it} = E_{it}^{M} + E_{it}^{S}, \tag{18}
$$

where the mean product-line component is

$$E_{it}^{M} = \frac{1}{|\mathcal{Y}_{it}|} \sum_{j \in \mathcal{Y}_{it}} E_{it}^{j}, \tag{19}$$

and the specialization term captures how inputs are tilted toward the plant's better products. The empirical approximation used in the paper is the product-code demeaned mean efficiency

$$\widehat{E}_{it}^{M} = \frac{1}{|\mathcal{Y}_{it}|} \sum_{j \in \mathcal{Y}_{it}} \left(E_{it}^{j} - \frac{1}{|\Lambda_{g(j),t}|} \sum_{k \in \Lambda_{g(j),t}} E_{it}^{k} \right). \tag{20}$$

- **Interpretation.** Mean product-level efficiency explains almost all of plant-level revenue efficiency, while specialization effects contribute only about 3% to 4% of its variance.
- **Why product-code demeaning?** It removes level differences across product codes due to different competition or market structure.

3.14 Core-competence technology

After ranking a plant's products from highest to lowest revenue efficiency, the within-plant efficiency ladder is summarized by

$$\widehat{E}_{it}^{r} - \widehat{E}_{it}^{1} = -H(r), \qquad r = 1, 2, \ldots, J_{it}, \qquad H'(r) > 0. \tag{21}$$

The paper estimates both linear and log-linear forms:

$$H(r) = b(r-1), \qquad H(r) = b \log r. \tag{22}$$

- **Interpretation.** The further a product is from the plant's core product, the lower its revenue efficiency.
- **Quantitative use.** These estimated ladders are then used to calculate how much plant-level efficiency would rise if the plant dropped its worst product.

3.15 Decomposing revenue efficiency into profitability, cost, and appeal

Revenue efficiency is decomposed as

$$E_{it}^{j} = P_{t}^{g(j)} + C_{it}^{j} + A_{it}^{j}, \tag{23}$$

where

$$P_{t}^{g(j)} = \frac{1}{|\Lambda_{g(j),t}|} \sum_{k \in \Lambda_{g(j),t}} E_{it}^{k} \tag{24}$$

is product-code profitability,

$$C_{it}^{j} = \widetilde{q}_{it}^{j} - \frac{1}{|\Lambda_{g(j),t}|} \sum_{k \in \Lambda_{g(j),t}} \widetilde{q}_{it}^{k} \tag{25}$$

is relative cost advantage, and

$$A_{it}^{j} = \widetilde{h}_{it}^{j} - \frac{1}{|\Lambda_{g(j),t}|} \sum_{k \in \Lambda_{g(j),t}} \widetilde{h}_{it}^{k} \tag{26}$$

is relative appeal advantage.

Each component is further split into plant averages and within-plant deviations:

$$X_{it}^{j} = \underbrace{\frac{1}{|\mathcal{Y}_{it}|} \sum_{k \in \mathcal{Y}_{it}} X_{it}^{k}}_{\text{across plant } (\overline{X}_{it})} + \underbrace{\left(X_{it}^{j} - \frac{1}{|\mathcal{Y}_{it}|} \sum_{k \in \mathcal{Y}_{it}} X_{it}^{k} \right)}_{\text{within plant } (\widehat{X}_{it}^{j})}. \tag{27}$$

- **Interpretation.** This decomposition separates general plant performance from a product's ranking within the plant's own internal ladder.
- **Why it matters.** A core-competence story is about *within-plant* rankings, not just about some plants being better on average than others.

3.16 Product-dropping regressions

The paper estimates probit models for whether a product disappears from a plant's product set in the next period. Schematically,

$$\Pr\Big(\text{Drop}_{i,j,t+1} = 1 \mid Z_{it}^{j}\Big) = \Phi\Big(\alpha + \beta Z_{it}^{j} + \text{year FE}\Big), \tag{28}$$

where Z_{it}^{j} is alternately total revenue efficiency E_{it}^{j}, product-code profitability, relative cost advantage, relative appeal advantage, or their within-plant components.

- **Interpretation.** Negative coefficients mean plants are less likely to keep weak products.
- **Economic point.** These regressions test whether plants behave *as if* they recognize internal productivity and appeal differences when choosing product scope.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The core identification problem is that product-level output differences reflect both product-line inputs and product-line productivity.
- The paper solves this using two pieces of structure:

  - product-level prices reveal marginal costs through demand and pricing first-order conditions;
  - homogeneous product-line technologies imply that input shares are proportional to marginal-cost-weighted output shares.

- This means the plant-product input allocation problem is identified before estimating the production function.

4.2 What makes the input-allocation theorem credible

- **Homogeneous technologies.** Because the production function is homogeneous of degree ϕ, cost minimization implies the same relative input mix across a plant's product lines, up to scale.
- **Connected substitutes.** This ensures invertibility of the pricing system and therefore recovery of marginal costs.
- **Observed product prices and quantities.** These are the essential extra data objects. Without them, the theorem cannot be used.
- **Full information for the firm.** Plants observe current product-line productivity when choosing inputs. This is why proxy-variable estimators based on unobserved productivity shocks are not compatible with the paper's structure.

4.3 Identification of demand

- The estimating equation is linear in observables once expressed in terms of log revenue shares.
- Price and within-nest shares are endogenous because product appeal h_{it}^j is unobserved and likely correlated with pricing.
- The first instrument uses input-price shocks in other industries that buy similar inputs.
- The second instrument uses the plant's own output mix to convert those cost shocks into additional product-level variation.
- The paper excludes input codes for which machinery is a major downstream consumer, so that the instrument is less likely to be contaminated by machinery-demand shocks themselves.

4.4 Identification of the production function

- Once input allocations are estimated, the paper treats $(\widehat{L}_{it}^j, \widehat{K}_{it}^j, \widehat{M}_{it}^j)$ as generated regressors and estimates the production function by nonlinear GMM.
- Lagged plant inputs are used as instruments because current productivity innovations are not observed at $t-1$.
- Capital is treated as predetermined, so current capital can instrument itself.
- The preferred specification uses current and lagged cost-shifter instruments to identify the materials elasticity rather than lagged materials alone.

4.5 Why the preferred production-function specification is column 4 of Table 4

- Specifications that use lagged materials as instruments imply a materials elasticity near 0.8 and returns to scale around 1.22, which look implausibly high in this setting.
- The preferred specification uses the external cost-shifter instruments and gives $\beta_L = 0.626$, $\beta_K = 0.236$, $\beta_M = 0.217$, and total returns to scale 1.078, close to constant returns.
- That specification also better fits the economic idea that non-joint production should not naturally generate large increasing returns.

4.6 Tests of assumption 2: identical production-function shape across product lines

- The paper estimates alternative specifications on single-product plants and single-industry plants and then tests whether output elasticities differ across 2-digit ASICC codes.
- The Wald tests do not reject equality of production functions across 2-digit codes in any of the reported specifications.
- This supports the key maintained assumption that product lines differ primarily through Hicks-neutral productivity rather than through different elasticities.

4.7 Interpretation of the dropping regressions

- These probits are not a clean causal estimate of one structural parameter. They are reduced-form evidence that plants act on internal heterogeneity.
- Their role is to show that the productivity and appeal measures are behaviorally meaningful, not just residuals with no economic content.

5 Tables and figures

5.1 Table 1: Plant-product-year summary statistics

Read:

- The typical plant-product observation is large in both revenue and quantity terms, but the dispersion is enormous: log revenues range from about 1.39 to 25.14.
- Roughly three-quarters of plant-product observations come from multiproduct plants, so internal allocation is a first-order issue in the raw data.
- About half the observations come from plants operating in more than one 2-digit industry, and about one-fifth are classified as potentially vertically integrated. This motivates the paper's careful sample restrictions.

TABLE 1
PLANT-PRODUCT-YEAR SUMMARY STATISTICS: MACHINERY, EQUIPMENT, AND PARTS (69,516 Observations)

Variable	Mean	Standard Deviation	Minimum	Maximum	Median
Log Revenue (r_{it}^j)	15.87	2.59	1.39	25.14	15.88
Log Quantity Sold (q_{it}^j)	7.65	4.17	−6.91	24.17	7.46
Log Prices (p_{it}^j)	8.23	3.6	−3.05	21.43	8.26
Log Quantity Produced (y_{it}^j)	7.78	4.18	−5.52	24.19	7.64
Multiproduct	.76	.43	0	1	1
Single Industry	.5	.5	0	1	0
Vertical Integration	.2	.4	0	1	0

NOTE.—Multiproduct, Single Industry, and Vertical Integration are all dummy variables. Single Industry refers to products produced by plants that produce only products belonging to ASICC codes 74–78. Vertical Integration refers to plants that produce output sets that I classify as potentially vertically integrated, using information on input use by single-product plants (see app. E1 for more details on the classification of potentially vertically integrated plants). Revenue is measured in nominal rupees, while quantity sold and quantity produced are measured in 5-digit ASICC code–specific units (kilograms, meters, number of units, etc.). Prices are measured as unit values for quantity sold, i.e., the ratio of nominal revenues in rupees, divided by quantity produced.

TABLE 2
PLANT-YEAR SUMMARY STATISTICS: MACHINERY, EQUIPMENT, AND PARTS (20,702 Observations)

Variable	Mean	Standard Deviation	Minimum	Maximum	Median
Log Labor (l_{it})	8.92	1.44	3.26	14.54	8.65
Log Capital Stock (k_{it})	15.15	2.25	−.09	23.48	14.94
Log Materials (m_{it})	6.91	3.29	−3.91	22.99	6.39
No. of varieties (J_{it})	1.55	1.31	1	28	1
Multiproduct	.26	.44	0	1	0

NOTE.—Summary statistics are reported only for plants that produce only products belonging to ASICC codes 74–78. Labor is measured in number of man-days worked; Capital is measured deflated rupees, using the capital deflator used in Allcott, Collard-Wexler, and O'Connell (2016); and Materials refers to measured materials expenditures (in current rupees) deflated by a plant-specific Cobb-Douglas price index. See app. E3 for more details on variable construction.

5.2 Table 2: Plant-year summary statistics

Read:

- The average plant produces 1.55 varieties, but the maximum is 28, so there is meaningful scope heterogeneity.
- Only about one-quarter of plants are multiproduct in the restricted machinery sample, but those plants are central because the whole paper is about within-plant allocation.
- The dispersion in labor, capital, and materials is large enough that plant-level heterogeneity remains substantial even before looking within plants.

5.3 Table 3: Demand estimates

Read:

- OLS produces a positive price coefficient, which is the wrong sign for demand. This is exactly what one expects when unobserved appeal is positively correlated with prices.
- IV flips the sign: the price coefficient becomes -0.220, and the within-nest share coefficient is 0.621.
- The paper interprets these estimates as implying an average own-price elasticity of about -1.6.
- First-stage coefficients on the cost-shifter instruments have intuitive signs, and the Sanderson–Windmeijer statistics are 16.35 and 10.07, suggesting usable instrument strength.

5.4 Table 4: Production-function estimates

Read:

- The preferred specification is column 4. It gives $\beta_L = 0.626$, $\beta_K = 0.236$, $\beta_M = 0.217$, and returns to scale 1.078.
- Columns 2 and 3 imply materials elasticities around 0.79 to 0.81 and overall returns to scale around 1.22. The paper views those estimates with skepticism.
- The message is that how one identifies the materials elasticity matters a lot. Using lagged materials alone appears to overstate scale economies.

TABLE 3
Demand Estimates (64,917 Observations)

	OLS	IV	p_{it}^{j}	$\mathrm{rs}_{it}^{jlg(j)}$
Estimates:				
p_{it}^{j}	.007	−.220		
	(.002)	(.116)		
$\mathrm{rs}_{it}^{jlg(j)}$	.946	.621		
	(.004)	(.306)		
First stage:				
$Z_{it}^{g(j)}$			.284	.136
			(.097)	(.041)
Z_{it}^{-jg}			.322	−.186
			(.198)	(.108)

Note.—Firm age controls, as well as dummies for product code, year, state, census status, rural locations, organization, ownership type, and the number of products, are included in all regressions. Sanderson-Windmeijer F-statistics are 16.35 and 10.07 for p_{it}^{jg} and for rs_{it}^{jlg}, respectively. Standard errors adjusted for two-way clustering by plant and product code are in parentheses. IV = instrumental variables.

TABLE 4
Production Function Estimates (3,620 Observations)

	OLS	GMM		
	(1)	(2)	(3)	(4)
β_L	.542	.331	.321	.626
	(.057)	(.192)	(.191)	(.261)
β_K	.223	.101	.097	.236
	(.040)	(.082)	(.081)	(.099)
β_M	.262	.790	.806	.217
	(.021)	(.191)	(.186)	(.352)
ρ^{74}		.757	.747	.842
		(.222)	(.197)	(.199)
ρ^{75}		.657	.661	.670
		(.082)	(.078)	(.068)
ρ^{76}		.651	.653	.623
		(.098)	(.104)	(.079)
ρ^{77}		.420	.422	.541
		(.062)	(.060)	(.087)
ρ^{78}		.194	.181	.569
		(.319)	(.265)	(.651)
$\beta_L + \beta_K + \beta_M$	1.027	1.222	1.224	1.078
	(.035)	(.084)	(.080)	(.113)
$(Z_{it}^{g(j)}, Z_{it-1}^{g(j)})$	...	Yes	No	Yes
$m_{i,t-1}$	...	Yes	Yes	No

Note.—OLS and GMM estimates of the production function (18) using an unbalanced panel of plant-product observations. All GMM specifications include $l_{i,t-1}$, k_{it}, $k_{i,t-1}$, and $y_{i,t-1}^{j}$ interacted with a 2-digit ASICC dummy, as instruments. Columns 2 and 3 also include $m_{i,t-1}$ as an instrument, while cols. 2 and 4 include $(Z_{it}^{g(j)}, Z_{it-1}^{g(j)})$, as defined by eq. (21). All specifications include a series of indicator variables for product code $g(j)$, as well as the total number of products produced by the plant. GMM specifications based on quasi differencing also include indicators for the lagged number of products produced by the plant, interacted with $\rho^{d(j)}$. Plant-level block bootstrapped standard errors are in parentheses (1,000 replications).

5.5 Table 5: Wald tests for equal production functions across 2-digit codes

Read:

- Across all reported specifications, the paper cannot reject equality of output elasticities across 2-digit ASICC codes.
- This is important because the entire input-allocation theorem is much cleaner when product lines share the same technology shape.
- The test is not definitive proof, but it gives empirical support for the maintained assumption used in the main estimation.

TABLE 5
Wald Tests: $F^{v} = F^{v'}$ (2,780 Observations)

	(1)	(2)	(3)	(4)	(5)	(6)
Wald test statistic	16.039	16.978	11.424	7.090	7.054	5.488
p-value	[.189]	[.150]	[.493]	[.852]	[.854]	[.940]
Single product	Yes	Yes	Yes	No	No	No
Single industry	No	No	No	Yes	Yes	Yes
$(Z_{it}^{g(j)}, Z_{it-1}^{g(j)})$	Yes	No	Yes	Yes	No	Yes
$m_{i,t-1}$	Yes	Yes	No	Yes	Yes	No

Note.—The above reports the Wald test statistics for a joint hypothesis test that $\beta_L^d = \beta_L^{d'}$, $\beta_K^d = \beta_K^{d'}$, and $\beta_M^d = \beta_M^{d'}$ for all (d, d') combinations in {74, 75, 76, 77, 78}, with the corresponding p-value for this test in brackets. Production function parameters and standard errors are reported in app. H1. Columns 1–3 use only single-product plants, while cols. 4–6 use only plants that produce within a single 2-digit ASICC code. All specifications use the standard set of instruments described in table 4, as well as the input price instruments $(Z_{it}^{g(j)}, Z_{it-1}^{g(j)})$ in cols. 1, 3, 4, and 6. In cols. 1, 2, 4, and 5, I also use lagged materials as an instrument.

TABLE 6
Multiproduct Plant Variance Decompositions
(11,933 Observations)

Type	Across	Within	Total
TFPQ:			
Variance	.682	.390	1.072
Percentage (%)	64	36	100
Appeal			
Variance	.666	.397	1.063
Percentage (%)	63	37	100

Note.—Only products produced by multiproduct plants are included in the sample. TFPQ corresponds to log quantity TFP, ω_{it}^{j}. Appeal corresponds to the estimated residual from eq. (20). All variables have been standardized to have mean zero and unit variance by 5-digit product code. See text for definition of "Across" and "Within."

5.6 Table 6: Variance decomposition of multiproduct-plant heterogeneity

Read:

- For TFPQ, across-plant variance is 0.682 and within-plant variance is 0.390, so the within-plant component is 36% of the total.
- For appeal, across-plant variance is 0.666 and within-plant variance is 0.397, so the within-plant component is 37% of the total.

- The main lesson is that within-plant heterogeneity is not a small residual object. It is quantitatively important, even though across-plant heterogeneity remains larger.

5.7 Table 7: Decile-rank comparison of core versus other products

Read:

- Plants with a top-decile product do *not* produce only other top-decile products.
- A plant whose core product is in the top decile of the TFPQ distribution produces other products whose average rank is only about 6.61 deciles. For appeal, the corresponding number is about 6.90.
- This is intuitive evidence for an internal efficiency ladder: being excellent at one product predicts being above average elsewhere, but far from equally strong everywhere.

TABLE 7
Decile Ranks: Core versus Other Products

	TFPQ Decile (1)	Appeal Decile (2)
2nd decile (10th–20th percentiles)	1.2985	1.2872
	(.0712)	(.0645)
Top variety in 3rd decile (20th–30th percentiles)	1.8710	1.7820
	(.1090)	(.1050)
Top variety in 4th decile (30th–40th percentiles)	2.3741	2.1396
	(.1163)	(.1315)
Top variety in 5th decile (40th–50th percentiles)	2.9809	2.6886
	(.1747)	(.1614)
Top variety in 6th decile (50th–60th percentiles)	3.4419	3.2403
	(.1616)	(.1659)
Top variety in 7th decile (60th–70th percentiles)	4.0611	3.7018
	(.1839)	(.2413)
Top variety in 8th decile (70th–80th percentiles)	4.7582	4.5523
	(.2051)	(.2344)
Top variety in 9th decile (80th–90th percentiles)	5.9277	5.5000
	(.2578)	(.2270)
Top variety in 10th decile (90th–100th percentiles)	6.6091	6.8969
	(.2560)	(.2191)
Observations	2,122	2,143

Note.—OLS regressions of a series of indicators for the decile rank of a plant's top product, according to product-code-year TFPQ ranking (col. 1) or product-code-year product appeal ranking (col. 2), on the decile rank of each particular variety within a plant (excluding the top-ranked variety). The sample consists of multiproduct plants that produce only product codes with at least 10 observations per product-code-year. Plants whose best-ranked product is in the bottom decile are dropped. Plant-level block bootstrapped standard errors are in parentheses (1,000 replications).

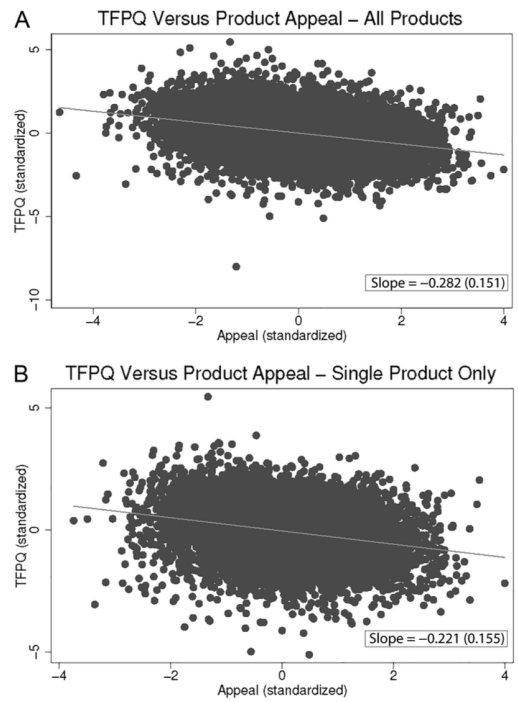

Fig. 1.—Negative correlation between TFPQ and product appeal. Appeal refers to η_a^j, the residual from equation (20), while TFPQ refers to ω_t^j in equation (18). Both variables have been standardized to have a mean of zero and unit variance within each product code. *A* includes all products, while *B* includes only single-product plants. Plant-level block bootstrapped standard errors are in parentheses (1,000 replications).

5.8 Figure 1: Negative correlation between TFPQ and product appeal

Read:

- The scatter plot shows a negative slope between standardized TFPQ and standardized appeal.
- For all products, the fitted slope is about -0.282; for single-product plants only, it is about -0.221.

- The interpretation is that physical productivity and demand-side appeal are not the same object. High-appeal products may embody unobserved quality, size, or durability that require more inputs, mechanically lowering measured TFPQ.
- This is why the paper shifts attention to *revenue efficiency* when it wants a plant-level measure that is comparable across products.

5.9 Table 8: Core-competence estimates

Read:

- The linear specification gives a coefficient of about -0.477 on $(r-1)$, implying a steep decline in efficiency as products move away from the core.
- The log-linear specification gives a coefficient of about -1.238 on $\log r$, implying even larger early drops from the best to the second- and third-best products.
- The nonparametric specification confirms the same pattern: the first few steps down the internal ladder are economically large.
- This table is the reduced-form estimate of the within-plant efficiency ladder that powers the product-dropping counterfactuals.

TABLE 8
WITHIN-PLANT HETEROGENEITY IN CORE-COMPETENCE MODELS

	Linear (1)	Log-Linear (2)	Nonparametric (3)
$r-1$	−.4773		
	(.1261)		
$\ln(r)$		−1.2378	
		(.3165)	
Indicator for $r=2$			−.9602
			(.2197)
Indicator for $r=3$			−1.4527
			(.3504)
Indicator for $r=4$			−1.7336
			(.4397)
Indicator for $r=5$			−1.9281
			(.5081)
Indicator for $r=6$			−2.0349
			(.5632)
Indicator for $r=7$			−2.0428
			(.6206)
Indicator for $r=8$			−2.1260
			(.6552)
Indicator for $r=9$			−2.3322
			(.6957)
Indicator for $r \geq 10$			−2.7367
			(.8775)
Observations	7,997	7,997	8,019

NOTE.—OLS regressions of $\hat{E}_{it}^{r} - \hat{E}_{it}^{1}$ on listed independent variables. Products with rank $j=1$ are excluded. Columns 1 and 2 exclude 22 plant-products that are ranked larger than 10 because of the rarity of these particular rankings. Plant-level block bootstrapped standard errors are in parentheses (1,000 replications).

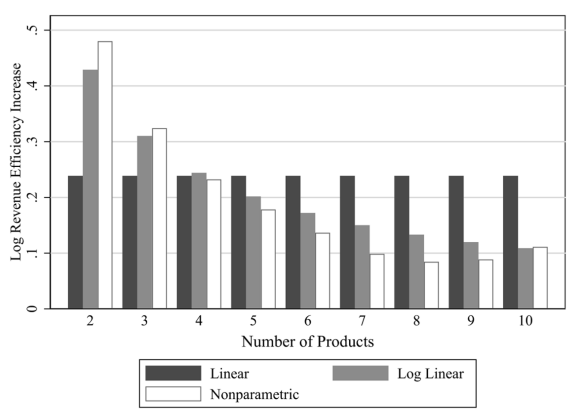

FIG. 2.—Plant-level revenue efficiency growth after removal of marginal varieties. The figure shows the implied log revenue efficiency growth from removing a plant's lowest-efficiency variety according to core-competence technology estimates in table 8. Log plant-level revenue efficiency is measured according to equation (30), with the y-axis denoting the change in plant-level log revenue efficiency after a plant's worst product is dropped.

5.10 Figure 2: Plant-level revenue-efficiency gains from removing the worst product

Read:

TABLE 9
PRODUCT-DROPPING PROBITS (3,472 Observations)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
E^j_{it}	−.0665							
	(.0218)							
$\bar{P}^g_t$		−.0318						−.0254
		(.0260)						(.0264)
$\bar{A}^j_{it}$			−.0403					−.0389
			(.0195)					(.0195)
$\bar{C}^j_{it}$				.0029				−.0078
				(.0174)				(.0207)
$\hat{P}^g_t$					−.0574			−.0896
					(.0220)			(.0256)
$\hat{A}^j_{it}$						−.0797		−.1277
						(.0188)		(.0289)
$\hat{C}^j_{it}$							.0088	−.0699
							(.0177)	(.0375)

NOTE.—Estimates of the marginal effects from a probit regression on an indicator variable for whether a plant drops a product in the following period. Definitions of all independent variables are described by eqq. (26), (32) and (33). E^j_{it}, and P^g_t have been standardized to have mean zero and unit variance over the entire sample, while C^j_{it} and A^j_{it} are standardized within a product-code-year. To guarantee that a product-dropping variable is capable of being constructed for all observations, the sample consists of multiproduct plants that I observe in the current as well as the subsequent period. All regressions include year fixed effects. Marginal effects are evaluated at the sample average for the covariates. Plant-level block bootstrapped standard errors are in parentheses (1,000 replications).

- The figure plots the implied increase in mean plant-level revenue efficiency after dropping the marginal variety.
- The gains are largest for two-product plants: depending on the specification, dropping the worst product raises mean plant-level revenue efficiency by roughly 22%, 47%, or just under 65%.
- For plants with many products, the gains are smaller but still meaningful, often around 10% or more.
- This is the central quantitative result of the paper's extensive-margin exercise.

5.11 Table 9: Product-dropping probits

Read:

- Column 1 shows that a one-standard-deviation fall in revenue efficiency raises the probability of dropping the product by about 6.65 percentage points.
- The full decomposition in column 8 shows that within-plant appeal shocks matter most: a one-standard-deviation decline in within-plant appeal advantage raises dropping probability by about 12.8 percentage points.
- Within-plant product-code profitability shocks also matter a lot, about 9.0 percentage points, and within-plant cost advantages matter too, about 7.0 percentage points.
- Across-plant averages are much less important than within-plant rankings. This is exactly what a core-competence model would predict.

6 Short summary

- **Conclusion.** The paper provides a general way to recover product-line input allocations and productivity for multiproduct producers using demand-side information, then shows that within-

plant productivity dispersion in Indian machinery manufacturing is large and behaviorally important.

- **Intuition.** Plants differ internally: a plant can be very good at one product and only moderately good at another. Prices and quantities reveal marginal costs; marginal costs reveal input allocations; input allocations reveal product-level productivity.
- **Empirical lesson.** Multiproduct producers should not be forced into single-product productivity frameworks. Once within-firm input allocation is modeled explicitly, one sees sizable internal heterogeneity and strong selection through product dropping.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Within-plant heterogeneity is large. It explains about one-third of variety-level TFPQ and appeal differences in the preferred estimates.
- **Identification contribution.** Under homogeneous product-line technologies and observable prices and quantities, input allocations are identified by marginal-cost-weighted output shares. This is the paper's main theoretical contribution.
- **Behavioral relevance.** Plants are significantly more likely to drop low-efficiency and low-appeal products, so the estimated heterogeneity matters for actual firm decisions.
- **Quantitative implication.** Extensive-margin changes in product scope can have large effects on plant-level revenue efficiency, especially for plants with only a few products.
- **Robustness.** The paper shows that its main message survives alternative competition assumptions, alternative treatments of transferability, and alternative ways of relaxing identical production functions.

7.2 Economic intuition

- A multiproduct plant is not a single undifferentiated technology. It is better interpreted as a collection of product lines with different comparative advantages.
- Product prices contain information about marginal costs and market power. Once demand is estimated, those prices tell us how inputs must have been allocated internally.
- Plants become "leaner and meaner" when they drop weak products. The evidence suggests that demand-side appeal differences are even more important than cost-side TFPQ differences in driving that pruning.

7.3 Takeaway

This paper gives a practical recipe for measuring within-firm productivity dispersion in multiproduct data sets that contain product-level prices and quantities but not product-level inputs. Its core

empirical message is simple and important: within-firm heterogeneity is real, large, and economically consequential. Plants do not just differ from other plants; they also differ sharply across their own product lines, and those internal differences help determine which products survive.